Sami Rana

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Education

Rensselaer Polytechnic Institute Troy, NY
Electrical Engineering/Computer and Systems Engineering B.S. *Aug. 2021 – May 2025*

Work Experience

Embedded Systems Engineer May. 2025 – Present
Deura Controls *Fairless Hills, PA*

- Co-founded startup developing intelligent sensor and control systems for industrial and enviromental applications
- Designed RS-485 multi-drop sensor network using STM32F0 microcontrollers, designing and integrating I^2C/SPI sensor drivers (LM75BD, SHT4x, BME280, AS7341, SCD41) for temperature, light, air quality, humidity, and pressure monitoring
- Established Git-based workflow and code reviews for embedded firmware bring-up, debugging, and validation

Electrical Engineer Co-op Jan. 2024 – Aug. 2024
Kulicke and Soffa Industries, Inc. *Fort Washington, PA*

- Built and debugged custom embedded Linux images for a SoC FPGA platform using Yocto/Bitbake; validated real-time data paths via AXI DMA and Linux IIO
- Redesigned PCB controlling XY axis linear motors, doubling efficiency, downsizing, & reducing costs
- Created LTspice simulations to optimize PCB design and perform comprehensive data analysis

Electrical Engineer Intern May 2022 – Aug. 2022
Kulicke and Soffa Industries, Inc. *Fort Washington, PA*

- Redesigned and specified a new upper-console pneumatic PCB, including I^2C-based airflow sensors (guide, diffuser, tensioner), to interface with master controller
- Developed and executed test scripts on a VxWorks RTOS host CPU in C to validate PCB performance using compressed air/nitrogen mechanical test rig

Research Experience

Undergraduate Research Assistant Sep. 2024 – Dec. 2024
Rensselaer Polytechnic Institute *Troy, NY*

- Analyzed DRAM timing parameters and memory controller behavior to support compression-aware memory architectures
- Simulated CPU, CXL controller, and DDR4 memory workloads to evaluate latency, bandwidth, and efficiency
- Developed multi-core simulations optimizing memory access for multithreaded workloads using DDR5 and shared buses

Projects/Leadership

Origami Art Installation | *Electrical Team Lead* January 2023 – December 2023

- Led Electrical development for user-interactive origami robot placed in engineering academic building
- Organized meetings and purchases
- Created schematic for active servos and microcontrollers

Formula Hybrid | *Ground Low Voltage (GLV) team lead* November 2021 – December 2023

- Led GLV subsystem development for a Formula Hybrid race car
- Designed and implemented PCBs and embedded firmware for vehicle subsystems
- Mentored first-year students on embedded systems fundamentals

Technical Skills

Operating Systems: Microsoft Windows 10, Microsoft Windows 11, Linux (Ubuntu, Debian, RHEL)
Languages/Libraries: Python, C/C++, x86/ARM assembly, Bash, HTML/CSS, VHDL, Verilog, MATLAB, OpenCV, TensorFlow, Numpy, CMSIS, ROS, freeRTOS
Embedded Systems: STM32 HAL/LL Drivers, RTOS threading and synchronization, DMA, ADC/DAC, UART, I2C, SPI, PWM, timers, interrupts, and GPIO control
CAD and Productivity Software: Altium Designer, LTSpice, VIM/Neovim, gem5, STMcube32IDE, Microsoft Office, Git, Docker, LabVIEW, Intel Quartus, Vivado, Yocto, WSL, Cadence Virtuoso